

Hospital Management System – Project Report

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AI/LLM Usage Declaration

AI/LLM tools were used only for **debugging assistance, documentation formatting, and layout/UI suggestions**.
All core logic, coding, design decisions, database models, and implementation were done manually by the author.
No auto-generated code or AI-written modules were directly submitted.

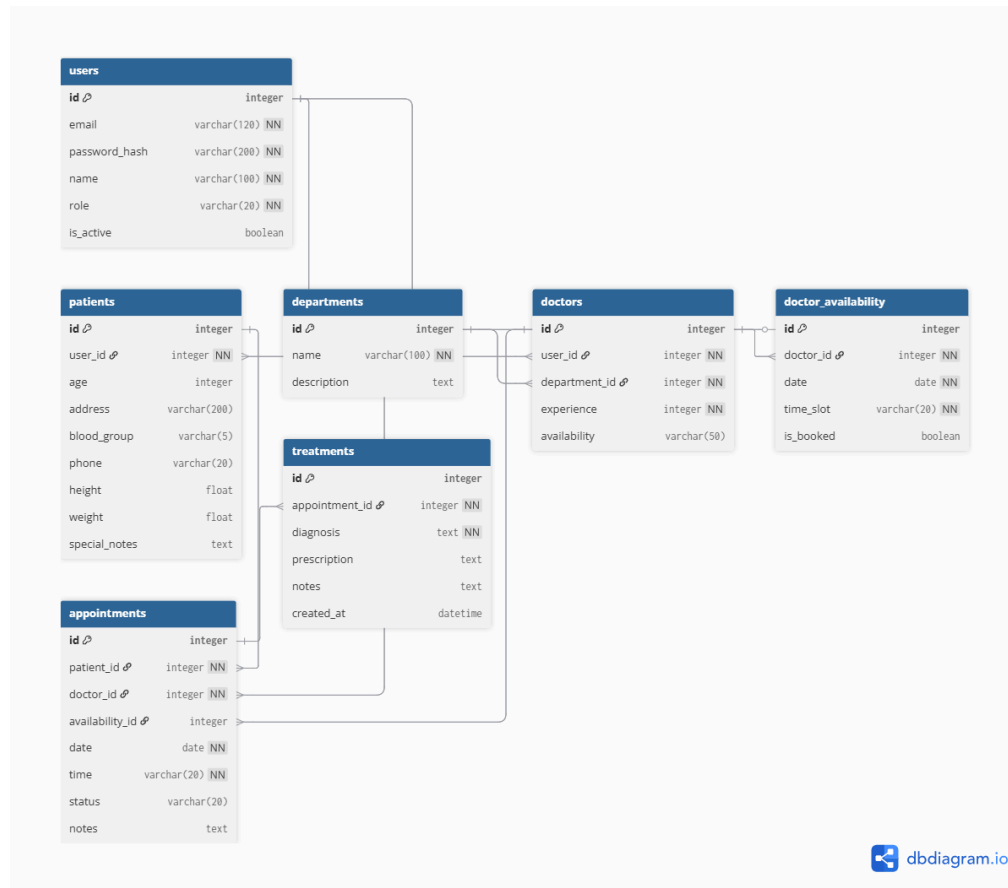
Project Description

The Hospital Management System (HMS) is a web-based application designed to streamline hospital operations by managing patients, doctors, appointments, and treatments. It addresses inefficiencies in manual or disconnected hospital record systems by providing role-based access for Admins, Doctors, and Patients, enabling appointment scheduling, treatment tracking, and administrative oversight through a centralized platform.

Technologies Used

Technology	Purpose
Flask	Backend web framework for routing and business logic
Flask-SQLAlchemy	ORM layer for database interactions with SQLite
Flask-Login	User session management and authentication
SQLite	Lightweight file-based relational database (<code>hms.db</code>)
Jinja2	Template engine for dynamic HTML rendering
Bootstrap	Responsive UI framework for front-end design
Chart.js	JavaScript library for analytics visualizations (optional)
Werkzeug	Password hashing and security utilities

Entity-Relationship Diagram



Workflow Example: Patient Booking Flow

1. Patient logs in → Dashboard shows departments
2. Clicks department → Views doctors in that specialization with slot counts
3. Clicks "Book Appointment" → Sees calendar of available slots (next 7 days)
4. Selects a slot → Appointment created with status "Booked", slot marked `is\_booked=True`
5. Doctor logs in → Sees appointment in "Upcoming" list
6. Doctor completes consultation → Marks appointment "Completed", adds treatment record
7. Patient views history → Sees diagnosis and prescription

Testing & Deployment

Local Setup:

```
# Install dependencies
pip install -r requirements.txt
```

```
# Initialize database (auto-runs on first start)
python app.py
```

Access:

- URL: `http://127.0.0.1:5000`
- Admin: `admin@hms.com` / `admin123`
- Doctor (sample): `ash@hms.com` / `doctor123`
- Patient (sample): `nick@hms.com` / `nick123`

Video Demonstration

Link:  mad-1 demo.mp4

Content Covered:

- Admin dashboard and management features
- Doctor availability setup and treatment workflow
- Patient registration, booking, and rescheduling
- Admin Analytics dashboard visualizations

Conclusion

The Hospital Management System successfully demonstrates a full-stack web application with role-based access control, dynamic scheduling, and data-driven analytics. The use of Flask's modular blueprint architecture ensures scalability, while SQLite and SQLAlchemy provide a lightweight yet robust data persistence layer. Future enhancements could include RESTful APIs for mobile integration, email notifications for appointment reminders, and advanced reporting features.